Analysis of Methods for Hiding Information in PDF Documents and Opportunities for Their Progress

Alexander V. Sergeev[1], Pavel B. Khorev[2]
[1, 2] Applied Mathematics Department
[1, 2] National Research University "MPEI"
[1, 2] Moscow, Russian Federation
[1]profitdre@gmail.com, [2]pbkh@yandex.ru

***Abstract*—Notwithstanding the popularity of multimedia files for steganography purposes, other files, whether binary data files, executables or text based files, can also be used to hide information. The widespread use of PDF files can make its use for this purpose an interesting and practical solution. In the article, an analysis was made of the structures of PDF documents, the use of steganography capabilities and their detection. A study of steganography algorithms from open sources was conducted and based on the analysis, an algorithm for detecting steganography was proposed. For this purpose, an integrated approach was developed.**

***Keywords*— steganography, analysis, PDF, data hiding.**

I. Introduction

Steganography is the art of covered or hidden writing, which is covert communication for secret messages to conceal the existence of a message from malicious attackers eavesdropping on a communication channel. The concept of information hiding is to conceal an important secret message in public information. The main part of the work in steganography is devoted to using images, video, sound and text as containers. However, textual steganography is the most difficult because of the lack of redundant information in text files, while much of the redundancy is present in images or sound files, which leads to high efficiency of using these files in steganography.

After the attacks on 9/11, many government intelligence agencies and industry experts believed the terrorist group al-Qaeda was using digital steganography to communicate among its terrorist cells. A group of researchers from the University of Michigan scanned websites, such as the auction listings on eBay, looking for potential terrorist communications hidden within the user uploaded images; no messages were found. [1]

In June 2010, the Federal Bureau of Investigation (FBI) arrested eleven Russian spies who were using digital steganographic technology to covertly communicate [2]. The spies were using these messages to “arrange meetings, cash drops, deliveries of laptops and further information exchanges”. This case shows that steganography is no longer solely used by child pornographers and financial fraudsters [2]. According to the National Institute of Justice (2010), “Newer steganography-encoding techniques are being rapidly developed, rendering the current detection tools ineffective” (para. 6) The rise of new digital steganography techniques can cause the effectiveness of the current of steganalysis techniques and programs to be questioned. The most popular document format in document management is PDF (Portable Document Format), which is of interest for further research. The PDF format created by Adobe Systems [3] for document sharing is a fixed layout format for presenting documents in a way that is independent of application software, hardware, or the operating system. Studies using these files as an information-hiding object are very few.

The structure of a PDF file consists of four parts: Header, Body, Cross-reference Table and Trailer. A PDF document consists of a set of objects that define the output data and functionality of a document, as shown in Fig. 1.

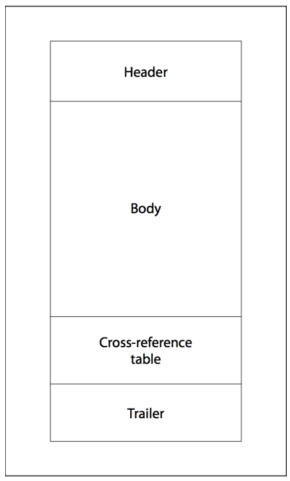

Fig.1. A sample of Pdf file structure

The PDF supports several basic data types: strings, arrays, dictionaries, streams, and objects: null, Boolean, integer, real number, name, string, array, dictionary and stream. Each of these types of objects and their feature in PDF format - strings (strings), arrays (arrays), dictionaries (distionaries), streams (streams) and objects (objects).

Very often, big data, images, and embedded fonts are compressed using the FlateDecode filter, which uses the same lossless compression technology used in the ZIP format. In PDF there are several ways to insert text data, for example, in the same Unicode encoding, octal character codes (\ 053), using a separate two-byte hex (<2B>) or even their sequence (<54776F20>).

II. Steganography in the Context of PDF

The PDF file stream contains various operators that can be used to display text, as well as to adjust the position of text within a PDF.

978-1-7281-5655-2/20/$31.00 ©2020 IEEE

Authorized licensed use limited to: Jain University. Downloaded on March 26,2026 at 09:42:51 UTC from IEEE Xplore. Restrictions apply.

The Tc operator is used to control a space between characters and extends to the entire text block.

The Tw operator is used to set a space between words. It works the same as the Tc operator, but only applies to the space character.

The TJ operator is used to display text strings in a PDF file and the space values that are used between string characters.

The Tm operator is a matrix of a coordinate system with specified x, y boundaries in a PDF document for further display of data in this matrix.

The Td operator - The coordinate operator of the beginning of the string inside the Tm matrix.

Discovered when analyzing existing ways to use steganography in PDF files: The coding algorithm with spaces, White Space Coding, use a regular and non-breaking space, which are visually indistinguishable, but have different codes [2]. In the TJ method (Text Justified) text alignment operators [3], data is embedded in the spacing between words and the use of text positioning operator [5, 6]. When aligning, the lengths of intervals take on a random character, which makes it possible to modify those using LSB (Least Significant Bit) methods.

To analyze the structure of the PDF document, an analysis of nine documents of this format was carried out. In order to get the numerical values stored in various PDF operators, a parsing of the document and its further decoding was carried out. [9-17]

In order to obtain the numerical values stored in various pdf operators, a syntax analysis procedure and its further decoding was carried out. Table 1 shows statistics on the number of operators in each document under consideration and the volume of the entire document.

TABLE I. STATISTICAL CHARACTERISTICS OF THE OPERATOR IN PDF

	1	2	3	4	5	6	7	8	9
TJ	802	1737	2910	4376	3636	1906	13235	3636	526
Tm	802	543	638	4376	503	94	855	3636	76
Tc	107	992	584	507	2673	67	0	342	307
Tw	0	383	318	0	2683	403	0	0	484
Td	0	917	833	0	2783	931	11958	0	435
Value (KB)	844	5309	17283	1410	1115	681	351	2623	3842

In the Tj operator, due to its features, several parameters may (but not necessarily) be present. Empty parameter values are also found in other operators. An analysis and appropriate sorting was carried out. Table 2 shows the number of parameters in the PDF documents analyzed.

TABLE II. RESULTS

	1	2	3	4	5	6	7	8	9
TJ	4550	5323	5866	28269	31681	1334	32475	22764	12788
Tm	1604	1074	1276	8752	1006	188	1710	7272	152
Tc	68	300	545	336	1747	30	0	223	182
Tw	0	203	291	0	1702	279	0	0	329
Td	0	880	833	0	2794	879	11931	0	429
TJ	4550	5323	5866	28269	31681	1334	32475	22764	12788

For further analysis, document No. 3 was selected [11], since it has all the specified operators and in the largest quantity, which is shown in Table 2.

Given the values of the Tj operator and using the algorithm [3], we apply the least significant bit (LSB) method. In the Tm operator, as well as in other Tc, Tw, Td operators, the last parameters are presented in (x, y) coordinates in a fractional form. According to one of the algorithms published in [5], full or partial replacement of decimal places in the values of operator parameters is used. Replacement occurs on the principle of converting the source text into a binary representation of 3 bits, which is then converted to a decimal number system as shown in the Table 3.

TABLE III. RESULTS

Portable Document Format

↓

010	100	000	110	111	101	110	010	011	101
000	110	000	101	100	010	011	011	000	110

Org.	Mod	Org.	Mod
000	1	100	5
001	2	101	6
010	3	110	7
011	4	111	8

↓

351	786	734	617	165	344	017

Considering that the total sum of the parameter values of the Tc, Tw, Td, Tm operators stored in the document does not exceed the number of Tj operators, the volume of information of the hidden message will be more than ~ 2945 bytes. Approximately 5866 bits (~ 733 byte) will be added to the Tj operator, assuming that the structure of the PDF document does not change and the number of Tj values remains unchanged. In article [1], an analysis was carried out and the dependencies of the embedded message in a PDF document were revealed. In the article, a frequency analysis of the data among the considered values of the operators from the Table 2 was conducted.

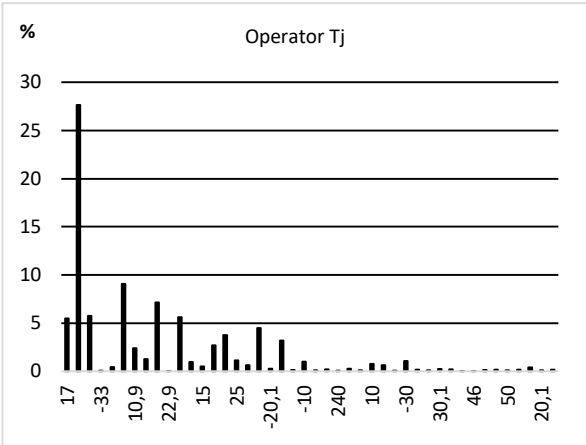

Fig.2. The obtained values of the operator TJ

From the graph shown in Fig. 2, it can be seen that the largest number of values, as shown in [5], lies in the interval [-16; 16], which is ~ 43% of the total number of parameter values.

Authorized licensed use limited to: Jain University. Downloaded on March 26,2026 at 09:42:51 UTC from IEEE Xplore. Restrictions apply.

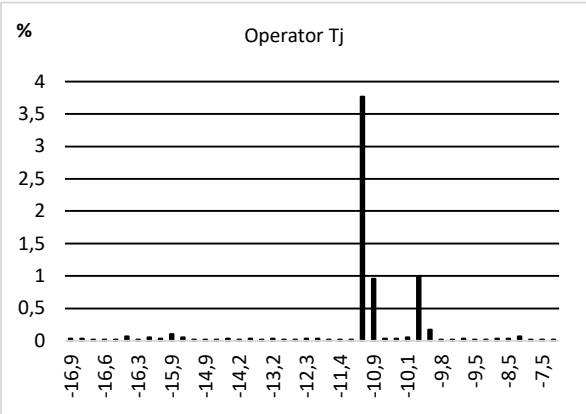

Fig.3. The obtained values of the TJ operator on the space (17;16]

The graph in Fig. 3 shows that the parameter values are fractional, which does not meet the necessary requirement for the implementation of information according to the algorithm given in [5]. When applying this algorithm, the volume of the embedded message will be reduced to ~ 35% of the total number of the Tj operator values. Not every PDF document has fractional values of the operator parameters, and during the analysis it was found that the PDF document designer is responsible for setting such characteristics.

In the article [5], other parameters of the operators were analyzed to determine the possibility of using the proposed algorithm for other operators, while fulfilling the necessary requirements of steganographic concealment

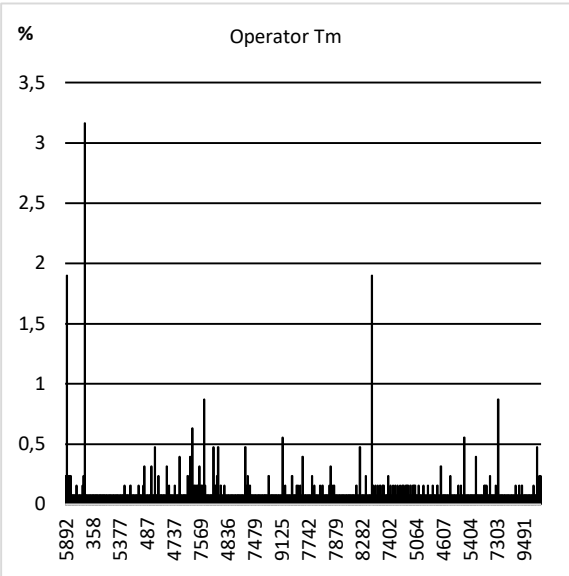

Fig.4. The obtained values of the operator Tm

Fig. 4 shows the values of the Tm operator. When analyzing 9 documents, a certain set of decimal places were found in the parameters (x) or (y) for the Tm(n+1) operator, which will be the same for Articles [7-8] did not consider the requirements for input data and the stability of the algorithm itself.

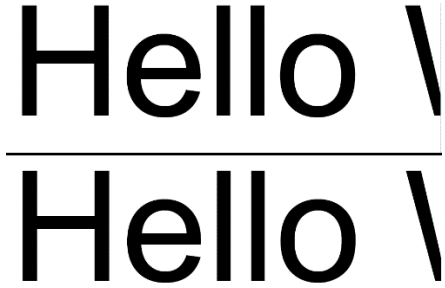

Fig.5. Operator mapping Tm. a) original display b) modified

Figure 5 shows the text with the original data, the first line [8.2495 450.8749] and replaced by [8.3517 450.8673] Tm, the second line. No visible differences were found, which corresponds to one of the properties of steganography.

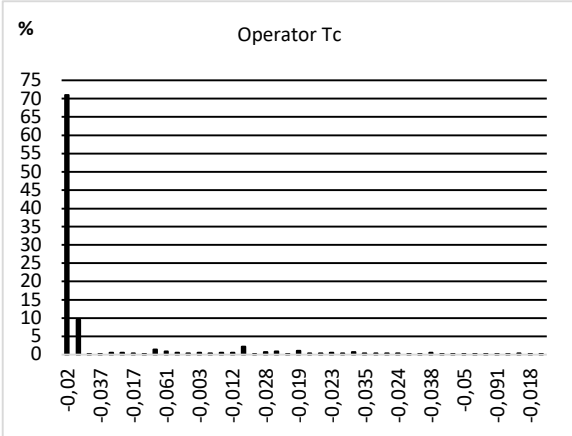

Fig.6. The obtained values of the operator Tc

Fig. 6 shows the values of the Tc operator. As you can see, ~ 80% are occupied by values of -0.02 and -0.025. The use of integer or fractional values of the Tc operator, rounded to the nearest hundredth, will be determined in the PDF document. Accordingly, if we use this operator for further processing, then the number of Tc operator parameters values will significantly decrease (for the document under consideration, to 109).

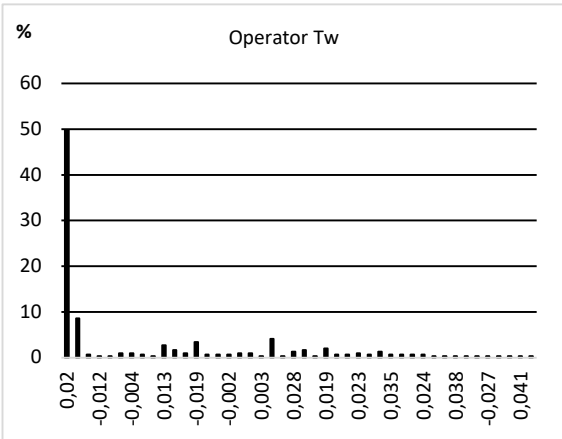

Fig.7. The obtained values of the operator Tw

Fig. 7 shows the parameters of the Tw operator. As you can see, ~ 50% are occupied by values of 0.02. The figure shows the text with the original data, the first line [0.0651 Tw] and replaced by [0.02 Tw] second line. No

Authorized licensed use limited to: Jain University. Downloaded on March 26,2026 at 09:42:51 UTC from IEEE Xplore. Restrictions apply.

visible differences were found, which corresponds to one of the properties of steganography.

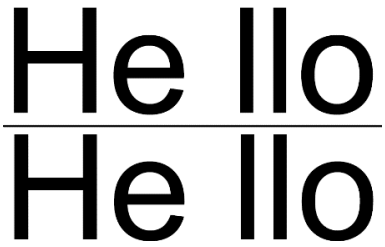

Fig.8. Operator mapping Tw. a) original display b) modified

When analyzing 9 documents, some set of values after the decimal place in the (x) or (y) parameters in the Td(n) operator as well as in the Td(n+1) operator was revealed (Fig. 9), as in the Td operator.

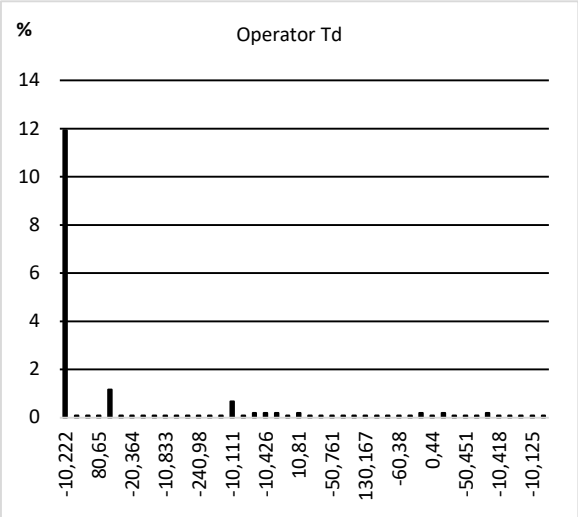

Fig.9. The obtained values of the operator Td

III. Conclusion

This paper According, an analysis of the results obtained makes it possible to detect data changes when hiding a message in a PDF document by mathematical analysis of the data.

As can be seen from the analysis, the Tj operator has a predominant number of parameter values, but due to the need to control entropy in the PDF document used as a steganographic container, the maximum volume of hidden message is reduced.

We propose to consider the method of mixed message embedding, which allows apply all the operators used in a PDF document.

The combined approach is proposed to be used as follows:

1. Statistics of user operations in the document are collected. The structure of the document is strictly regulated the operators follow in strict order.

2. Data array created from the value of Tj operator and an array of decimal places from the operators Tm, Tc, Tw, and Td.

3. Revealed allowable to implement messaging.

The hidden message evenly distributed among all operators of the container document, which replaced by the corresponding algorithms.

Thus, we increase the size of hidden data due to the large numbers of PDF operators used or reduce probability of detection of hidden information in the document.

References

[1] James C. Judge, "Steganography: Past, Present, Future", SANS Institute: InstituteInfoSec ReadingRoom, 2001

[2] Kelly Jackson Higgins, "Busted Alleged Russian Spies Used Steganography to Conceal Communications", Dark Reading, 2010, in press.

[3] ISO 32000-1 "Document management — Portable document format — Part 1", 2008 (references)

[4] I-Shi Lee, Wen-Hsiang Tsai, “A new approach to covert communication via PDF files”, Signal Processing, Vol. 90, 557-565, 2010.

[5] Zhong S., Cheng X., Chen T. "Data hiding in a kind of pdf texts for secret communicationl", International Journal of Network Security, 2007. - Vol. 4(1), pp. 17–26.

[6] Ndoundam R., Ekodeck S. G., "PDF steganography based on Chinese Remainder Theorem," J. Inf. Secur. Appl., vol. 29, pp. 1–15, 2016

[7] J. T. Wang and W. H. Tsai, “Data hiding in PDF files and applications by imperceptible modifications of PDF object parameters,” Conf. on Computer Vision, Graphics & Image Proc., Yilan, Taiwan, Aug. 24-26, 2008

[8] Y. C. Lai and W. H. Tsai, “Covert Communication via PDF Files by New Data Hiding Techniques”, In: Proc. of Conf. on Computer Vision, Graphics and Image Processing, 2009.

[9] JSC Advanced Monitoring, "Report of the Monitoring Center for the second half of 2017 Monitoring Center CJSC", in press, 2018

[10] Nikolai Samodaev, "Improving the information security system using the active protection program", Ernst & Young, in press, 2016

[11] Nikolai Samodaev, "Path to cyber resistance: prediction, protection, response", Ernst & Young [Digests 20th EY international research in information security Russia, 2016].

[12] The Financial Sector Computer Emergency Response Team (FinCERT) of the Bank of Russia, "Report of the Center for Monitoring and Responding to Computer Attacks in the Credit and Financial Sphere of the Central Directorate of Information Security and Protection of the Bank of Russia.", in press, 2016

[13] InfoWatch analytical center, "Global research of confidential information leaks in 2017", in press, 2018.

[14] Kaspersky Lab, Study of current trends in the field of information security business "Kaspersky Lab", in press, 2019.

[15] EFSOL, "It security audit report", in press, 2016.

[16] SearchInform, "Information Security Circuit", in press, 2015.

[17] Positive technologies, "Attacks to corporate wi-fi", in press, 2017.

Authorized licensed use limited to: Jain University. Downloaded on March 26,2026 at 09:42:51 UTC from IEEE Xplore. Restrictions apply.